In the above program, ios flag is set to **fixed** type; it prints the floating point number in fixed notation. So, the output will be, 0.10

cout.setf(ios::scientific);

cout << setprecision(2) << 0.1;

In the above statements, ios flag is set to **scientific type;** it will print the floating point number in scientific notation. So, the output will be, 1.00e-001

9.15 Expression

An expression is a combination of operators, constants and variables arranged as per the rules of C++. It may also include function calls which return values. (Functions will be learnt in upcoming chapters).

An expression may consist of one or more operands, and zero or more operators to produce a value. In C++, there are seven types of expressions, and they are:

(i) Constant Expression (ii) Integer Expression
(iii) Floating Expression (iv) Relational Expression
(v) Logical Expression (vi) Bitwise Expression
(vii) Pointer Expression

SN	Expression	Description	Example
1	Constant Expression	Constant expression consist only constant values	int num=100;
2	Integer Expression	The combination of integer and character values and/or variables with simple arithmetic operators to produce integer results.	sum=num1+num2; avg=sum/5;
3	Float Expression	The combination of floating point values and/or variables with simple arithmetic operators to produce floating point results.	Area=3.14*r*r;
4	Relational Expression	The combination of values and/or variables with relational operators to produce bool(true means 1 or false means 0) values as results.	x>y; a+b==c+d;
5	Logical Expression	The combination of values and/or variables with Logical operators to produce bool values as results.	(a>b)&& (c==10);
6	Bitwise Expression	The combination of values and/or variables with Bitwise operators.	x>>3; a<<2;
7	Pointer Expression	A Pointer is a variable that holds a memory address. Pointer variables are declared using (*) symbol.	int *ptr;

Table 9.10 : Types of Expressions

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